THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ray of light makes an angle of 30° with the normal to a plane mirror. The angle of reflection is: 1

(a) 30° **(b)** 150° **(c)** 60° **(d)** 40°

2. Light travels in: 1

(a) straight lines in a uniform medium **(b)** curves always
(c) circles **(d)** zig-zag always

3. The incident ray, reflected ray and normal lie in: 1

(a) the same plane **(b)** different planes **(c)** parallel planes **(d)** no plane

4. Lateral inversion means: 1

(a) bigger image **(b)** left appears as right
(c) smaller image **(d)** upside down image

5. A convex mirror makes light rays: 1

(a) stop **(b)** converge **(c)** diverge **(d)** stay parallel

6. Dentists and ENT specialists use ___ mirrors to get enlarged images. 1

(a) convex **(b)** plane **(c)** cylindrical **(d)** concave

7. A lens is a piece of transparent material with: 1

(a) a flat surface only **(b)** a rough surface only
(c) curved surface(s) **(d)** no surface

8. The image from a concave lens is always: 1

(a) erect and enlarged **(b)** inverted and diminished
(c) erect and diminished **(d)** inverted and enlarged

9. Corrective glasses for nearsighted eyes use a: 1

(a) plane mirror **(b)** convex mirror **(c)** convex lens **(d)** concave lens

10. A woman walks towards a large concave mirror. She sees her inverted image get bigger and then: 1

(a) a plane image **(b)** an erect, magnified image
(c) an inverted smaller image **(d)** no image

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. State the two laws of reflection. 2

12. What is a normal in ray diagrams? 2

13. If the angle of incidence is 35°, what is the angle of reflection and the angle between incident and reflected rays? 2

14. Name three characteristics of an image in a plane mirror. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A ray is incident on a plane mirror at 25° to the normal. Draw (describe) the reflected ray and find the angle of reflection and the angle between the reflected ray and the mirror. 3

16. Describe the activity to verify the laws of reflection. 3

17. What happens when a ray falls along the normal? Draw and explain. 3

OR

Compare the images formed by plane, concave and convex mirrors. 3

18. Explain how the distance of an object from a concave mirror changes the image. 3

OR

Why do vehicles use convex mirrors but dentists use concave mirrors? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. State and explain the laws of reflection. Describe an activity to verify them. 5

OR

Describe the images formed by a concave mirror when the object is at small distance and at large distance, and give uses. 5

20. Explain the characteristics of the image in a convex mirror and give two applications. 5

OR

Compare plane, concave and convex mirrors in a table on shape, image, lateral inversion and use. 5

END OF QUESTION PAPER — Total: 40 Marks

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